

REGENSBURG OBERHUB, Germany

WMO: 107760

Lat: 49.0424N

Lon: 12.1020E

Elev: 1216

StdP: 14.06

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	12.0	1.2	6.1	9.1	6.3	7.9	13.7	24.4	42.4	21.3	40.3	4.7	0	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	85.4	65.7	81.9	64.2	78.9	63.1	68.1	80.3	66.5	77.3	65.0	74.9	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	94.5	71.0	62.8	89.4	69.6	61.4	85.1	68.4	33.0	81.0	31.7	77.4	30.6	74.9	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	14.5	12.3	DB	3.5	92.0	8.1	4.1	-2.3	95.0	-7.0	97.3	-11.6	99.6	-17.4	102.6	(4)
(5)			WB	3.1	71.9	8.0	1.6	-2.6	73.0	-7.3	73.9	-11.7	74.8	-17.5	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	29.6	30.1	39.7	46.9	55.9	61.3	65.6	65.4	57.9	48.4	37.3	32.4	(6)
(7)		DBStd	14.69	9.46	8.71	7.02	6.34	6.41	6.28	5.58	5.90	5.46	6.38	6.08	7.05	(7)
(8)		HDD50	2709	633	558	324	133	23	1	0	0	3	108	382	544	(8)
(9)		HDD65	6542	1098	978	783	543	285	146	61	70	222	516	831	1009	(9)
(10)		CDD50	1853	0	0	6	40	206	340	485	477	241	57	1	0	(10)
(11)		CDD65	210	0	0	0	0	2	35	81	82	10	0	0	0	(11)
(12)		CDH74	2101	0	0	0	3	66	352	855	739	86	0	0	0	(12)
(13)		CDH80	519	0	0	0	0	1	65	237	211	5	0	0	0	(13)
(14)	Wind	WSAvg	5.0	6.0	5.6	5.9	5.7	4.8	4.5	4.3	4.1	4.2	4.3	5.1	5.6	(14)
(15)	Precipitation	PrecAvg	25.0	1.8	1.4	1.6	1.5	2.4	3.1	2.8	2.9	1.9	1.9	2.0	1.9	(15)
(16)		PrecMax	33.3	3.8	5.7	5.4	3.8	4.5	6.3	6.1	5.7	4.1	5.9	4.5	4.4	(16)
(17)		PrecMin	17.7	0.7	0.3	0.1	0.4	0.6	0.6	0.9	0.9	0.4	0.2	0.3	0.2	(17)
(18)		PrecStd	3.9	0.7	1.0	1.0	0.9	1.0	1.3	1.4	1.1	0.9	1.5	0.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.2	53.7	66.1	72.8	79.2	85.9	90.7	90.0	80.2	70.4	55.3	49.8	(19)
MCWB			44.6	46.8	52.1	54.6	60.4	65.9	67.7	66.4	63.5	58.4	50.3	46.7	(20)	
2%		DB	46.3	47.2	59.1	67.7	75.7	81.4	85.5	85.7	76.6	64.7	51.8	46.9	(21)	
		MCWB	42.5	42.3	47.8	52.9	58.3	63.9	66.2	65.5	63.0	56.0	49.2	44.5	(22)	
5%		DB	43.5	43.5	53.8	63.8	73.0	78.1	82.2	81.6	72.6	60.9	48.1	44.6	(23)	
		MCWB	41.1	40.0	45.5	50.6	57.0	62.7	64.5	64.0	61.4	54.5	45.0	42.8	(24)	
10%		DB	40.7	40.5	49.9	59.8	69.9	74.5	78.9	77.9	68.7	57.6	45.5	41.9	(25)	
		MCWB	38.8	37.7	43.9	49.3	55.6	61.1	63.5	62.9	58.6	52.8	43.1	40.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	47.1	53.5	56.6	63.6	69.2	70.8	69.5	66.5	61.1	51.8	47.6	(27)
MCDWB			49.2	53.0	65.0	68.8	72.8	80.3	83.4	84.2	75.9	67.6	53.6	49.1	(28)	
2%		WB	43.5	43.3	49.2	54.1	60.6	66.9	68.3	67.3	64.6	57.7	49.1	45.0	(29)	
		MCDWB	45.6	46.3	55.8	65.0	69.9	76.8	80.4	80.5	73.3	62.6	51.6	46.5	(30)	
5%		WB	41.2	40.4	46.7	52.0	58.8	64.7	66.7	65.7	62.7	55.3	45.5	42.8	(31)	
		MCDWB	43.2	42.7	51.9	61.4	68.5	74.3	77.6	76.7	69.5	59.0	47.4	44.6	(32)	
10%		WB	38.6	37.9	44.8	50.1	57.1	62.6	65.1	64.4	60.6	53.4	43.4	40.3	(33)	
		MCDWB	40.6	40.2	49.7	57.8	67.9	71.7	75.2	73.7	66.6	56.9	45.0	41.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.4	11.4	14.8	18.6	20.1	18.9	20.8	20.2	18.0	14.5	8.9	7.3	(35)
MCDBR			10.8	13.5	23.2	27.0	26.5	25.0	27.5	26.9	23.6	19.3	12.5	9.6	(36)	
MCWBR			8.3	9.2	12.7	12.9	11.1	10.0	10.5	9.7	11.2	11.2	9.0	8.2	(37)	
5% WB		MCDWB	10.2	12.9	19.9	24.0	22.2	22.2	22.9	22.2	20.2	17.3	10.9	9.3	(38)	
		MCWBR	8.3	9.5	11.8	12.1	10.5	10.0	10.0	9.1	10.4	11.1	8.7	8.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.325	0.362	0.396	0.403	0.405	0.413	0.407	0.380	0.354	0.328	0.300	(40)	
taud		2.417	2.367	2.320	2.260	2.272	2.296	2.269	2.307	2.358	2.421	2.454	2.439	(41)		
Ebn at Noon		240	260	268	270	272	272	267	263	259	246	228	226	(42)		
Edh at Noon		21	27	34	39	41	40	41	37	32	26	20	18	(43)		
(44)	All-Sky Solar Radiation	RadAvg	293	572	913	1355	1586	1753	1675	1468	1053	641	318	227	(44)	
(45)		RadStd	49	83	118	176	184	148	160	146	138	75	37	28	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (60 neighbors)		+0.70	N/A	N/A	N/A	N/A	+0.63	N/A	N/A	+99	+36

Nomenclature: See separate page